Endogenous Growth Cheatsheet

Lecture 4: AK Model and Romer Expanding Varieties

Macroeconomics I – Exam Reference

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1 Notation and Definitions

Core Variables

K = aggregate capital stock
L = labor (population, fixed or growing at rate n)
$k = K/L$ = capital per worker
$c = C/L$ = consumption per worker
δ = depreciation rate
σ = inverse elasticity of intertemporal substitution
Y = aggregate output
A = technology / TFP
$y = Y/L$ = output per worker
s = savings rate (exogenous)
ρ = discount rate
n = population growth rate

Model-Specific Notation

AK and Learning-by-Doing:

A = aggregate productivity level
η = externality parameter (learning-by-doing)
$\gamma_x = \dot{x}/x$ = growth rate of variable x
α = capital share in individual firm production
$\bar{A} = A_0 K^\eta$ = firm-level productivity

Romer Expanding Varieties:

M_t = measure of intermediate goods (varieties)
L_Y = labor in final good production
π = per-period monopoly profit
λ = research productivity
w = wage rate
x_i = quantity of intermediate good i
L_R = labor in R&D
V = present value of firm / patent
r = interest rate
ϕ = spillover parameter (Jones model, $\phi < 1$)

Key Economic Concepts

- **Balanced Growth Path (BGP):** All variables grow at constant rates (possibly zero).
- **Endogenous growth:** Long-run positive growth without exogenous TFP process.
- **Scale effects:** Growth rate increases with economy size L.
- **Non-rival good:** Use by one agent does not preclude use by another (e.g., knowledge).
- **Monopolistic competition:** Firms are price-setters for their variety, price-takers for aggregates.

2 AK Model: Basic Setup (Exogenous Savings)

Production Function

The AK model assumes a linear relationship between capital and output, eliminating diminishing returns:

$$y = f(k) = Ak$$

This is equivalent to a Solow model with $\alpha = 1$, so $f_{kk} = 0$. Capital per worker $k = K/L$.

Capital Accumulation

With exogenous savings rate s and population growth n:

$$\dot{K} = sY - \delta K = sAK - \delta K$$

The growth rate of capital is therefore:

$$\gamma_K = \frac{\dot{K}}{K} = sA - \delta$$

Since $Y = AK$ is proportional to K, output and consumption grow at the same rate:

$$\gamma_Y = \gamma_K = \gamma_C = sA - \delta$$

Growth Rate of Per-Capita Variables

Since $k = K/L$ and L grows at rate n:

$$\gamma_k = \gamma_K - n = sA - \delta - n$$

$$\boxed{\gamma_k = \gamma_y = \gamma_c = sA - \delta - n} \qquad \text{(AK Growth)}$$

Key Properties of the AK Model

- **No diminishing returns to capital:** $f_{kk} = 0$, so the marginal product of capital A is constant.
- **No transitional dynamics:** The economy is always on its BGP. There is no convergence period – if $sA > \delta + n$, the economy grows immediately and indefinitely.
- **No convergence:** Unlike Solow, there is no tendency for poor countries to catch up. Different countries with the same s grow at the same rate, but level differences persist.
- **Savings rate affects growth:** An increase in s permanently raises γ_k. This is a sharp departure from Solow.
- **Positive growth condition:** $sA > \delta + n$ is required for sustained per-capita growth.

AK vs. Solow Contrast

Feature	Solow	AK
Returns to capital	Diminishing	Constant
Transitional dynamics	Yes (converges to SS)	No
Savings $\uparrow$	Level effect only	Permanent growth effect
Convergence	Conditional convergence	No convergence

3 AK-Ramsey Model (Endogenous Savings, Euler Equation)

Setup

The household maximizes lifetime utility under CRRA:

$$U(0) = \int_0^\infty \frac{c^{1-\sigma} - 1}{1-\sigma} e^{(n-\rho)t}\, dt$$

The capital law of motion (using AK technology and competitive factor markets):

$$\dot{k} = Ak - c - (\delta + n)k = (A - \delta - n)k - c$$

In equilibrium, the rental rate of capital equals $r_t = A - \delta$ (since $F_K = A$ and capital is rented at its marginal product).

Current-Value Hamiltonian

Step 1: Set up the Hamiltonian.
The control variable is c, the state variable is k, and the shadow price is λ:

$$H^{CV} = \frac{c^{1-\sigma}}{1-\sigma} + \lambda\big[(A - \delta - n)k - c\big]$$

Step 2: First-order condition with respect to control c.

$$\frac{\partial H^{CV}}{\partial c} = 0 \implies c^{-\sigma} = \lambda$$

Step 3: Costate equation (FOC with respect to state k).

$$\frac{\partial H^{CV}}{\partial k} = \rho\lambda - \dot{\lambda} \implies \lambda(A - \delta - n) = \rho\lambda - \dot{\lambda} \implies \frac{\dot{\lambda}}{\lambda} = \rho - (A - \delta - n)$$

Step 4: Derive the Euler equation.
Differentiate $c^{-\sigma} = \lambda$ with respect to time:

$$-\sigma c^{-\sigma-1}\dot{c} = \dot{\lambda} \implies \frac{\dot{\lambda}}{\lambda} = -\sigma\frac{\dot{c}}{c}$$

Substitute from Step 3:

$$-\sigma\frac{\dot{c}}{c} = \rho - (A - \delta - n) \implies \frac{\dot{c}}{c} = \frac{(A - \delta - n) - \rho}{\sigma} = \frac{A - \delta - n - \rho}{\sigma}$$

Note: Since $r_t = A - \delta$ is the market interest rate, this is the standard Euler equation $\gamma_c = (r - \rho)/\sigma$ evaluated at $r = A - \delta$, subtracting n to get per-capita growth.

$$\boxed{\gamma_c = \frac{\dot{c}}{c} = \frac{A - \delta - \rho}{\sigma}} \qquad \text{(AK-Ramsey Euler)}$$

(This is the per-capita growth rate; if population is growing at n, replace ρ with $\rho - n$ in the utility or adjust the constraint accordingly.)

Transversality Condition (TVC)

$$\lim_{t\to\infty} k_t e^{(n-r)t} = 0$$

This imposes $\rho > (1-\sigma)\gamma_c$, ensuring the household does not over-accumulate.

Key Properties of AK-Ramsey

- Growth is positive iff $A > \delta + \rho$ (for the simplest case with $n = 0$).
- The growth rate is **constant** immediately – no transitional dynamics.
- Higher patience (lower ρ) or higher A raises growth permanently.
- Higher σ (less willingness to substitute intertemporally) lowers growth.
- The savings rate s is now determined endogenously; it will be constant on the BGP.
- **On the BGP:** Since γ_c and γ_k are both equal to $(A - \delta - \rho)/\sigma$, the capital-consumption ratio k/c is constant.

4 Learning-by-Doing Externality (Arrow-Frankel)

Individual Firm Technology

Each firm j in a continuum operates:

$$Y_j = \bar{A} K_j^{\alpha} L_j^{1-\alpha}, \qquad \bar{A} = A_0 K^{\eta}, \quad \eta > 0$$

where $K = \int_i K_i$ is the aggregate capital stock. Each firm takes $\bar{A}$ as given (external).

Aggregate Production Function

Step 1: Substitute $\bar{A} = A_0 K^{\eta}$ into the firm production function.
Step 2: Use market clearing: $\int_i K_i = K$ and $\int_i L_i = L$. Since goods are perfect substitutes, $Y = \int_i Y_i$.
Step 3: Aggregate over a unit measure of symmetric firms:

$$Y = A_0 K^{\eta} \cdot K^{\alpha} L^{1-\alpha} = A_0 K^{\eta+\alpha} L^{1-\alpha}$$

Step 4: Define $A \equiv A_0 L^{1-\alpha}$ for convenience. Then:

$$Y = A K^{\eta+\alpha}$$

$$\boxed{Y = A_0 K^{\eta+\alpha} L^{1-\alpha}} \qquad \text{(Aggregate Arrow-Frankel)}$$

Growth Scenarios

The growth rate of capital with exogenous savings is:

$$\gamma_K = sAK^{\eta+\alpha-1} - \delta$$

Condition	Regime	Outcome
$1-\alpha-\eta > 0$	Diminishing returns	Solow-like: converges to $K^* = (sA/\delta)^{1/(1-\alpha-\eta)}$
$1-\alpha-\eta = 0$	Knife-edge (AK case)	Sustained endogenous growth
$1-\alpha-\eta < 0$	Increasing returns	Explosive growth

The AK Case: $\eta = 1 - \alpha$

When the externality exactly offsets diminishing returns, $\eta + \alpha = 1$:

$$Y = A_0 K L^{1-\alpha} = Ak \qquad \text{(with } L = \bar{L} = 1 \text{ normalized)}$$

where $A \equiv A_0 \bar{L}^{1-\alpha}$. This is the AK model with:

$$\gamma_k = sA - \delta$$

Private vs. Social Returns to Capital

Private return: Each firm optimizes taking $\bar{A}$ as given:

$$R_t = \frac{\partial Y_j}{\partial K_j} = \alpha \bar{A} K_j^{\alpha-1} L_j^{1-\alpha} = \alpha A \quad \Rightarrow \quad r_t = \alpha A - \delta$$

(using $K_j = K$, $L_j = L = 1$ in symmetric equilibrium)

Social return: A planner internalizes the effect of K_j on $\bar{A}$:

$$R_t^{SP} = \frac{\partial Y}{\partial K} = (\alpha + \eta) A_0 K^{\alpha+\eta-1} L^{1-\alpha} = A \quad \Rightarrow \quad r_t^{SP} = A - \delta$$

(using $\alpha + \eta = 1$ in the AK case)

Market vs. Planner Euler Equations

	Euler Equation	Growth Rate
Market	$\gamma_c = \dfrac{r_t - \rho}{\sigma}$	$\boxed{\gamma_c^{mkt} = \dfrac{\alpha A - \delta - \rho}{\sigma}}$
Planner	$\gamma_c^{SP} = \dfrac{r_t^{SP} - \rho}{\sigma}$	$\boxed{\gamma_c^{SP} = \dfrac{A - \delta - \rho}{\sigma}}$

Since $\alpha \in (0, 1)$, we have $\alpha A < A$, so $\gamma_c^{mkt} < \gamma_c^{SP}$.

Inefficiency and Optimal Subsidy

- **Source of inefficiency:** The externality $\eta > 0$ means the social marginal product of capital (A) exceeds the private marginal product (αA). Firms under-accumulate capital relative to the social optimum.
- **Direction:** Too little saving and investment in equilibrium.
- **Optimal subsidy:** A subsidy s^* on capital rental that closes the gap between private and social returns. The firm's effective cost of capital becomes $\alpha A - s^*$, and we need this to equal the social return A. This requires:

$$\boxed{s^* = \frac{1 - \alpha}{\alpha}}$$

(expressed as a fraction of the rental cost, so the after-subsidy private return equals the social return)

Intuition: Externality Direction

If $\eta > 0$ (positive externality): social MPK $>$ private MPK $\Rightarrow$ market underinvests. Planner grows faster. Optimal policy: subsidize capital.

If $\eta < 0$ (negative externality): social MPK $<$ private MPK $\Rightarrow$ market overinvests. Planner grows *slower*. Optimal policy: tax capital.

5 Romer Expanding Varieties: Market Equilibrium

Setup and Production Structure

There are two sectors:

- **Final good:** Competitive firm using labor L_Y and a measure M_t of intermediates x_i.
- **Intermediate goods:** Each variety i produced by a monopolist with perpetual patent.

No capital in this economy. Household savings flow into R&D (firm equity).

Final Good Production

$$Y_t = \frac{L_Y^{1-\alpha}}{\alpha} \int_0^{M_t} x_i^\alpha \, di, \qquad \alpha \in (0,1)$$

The $1/\alpha$ is a normalization. The final good firm is competitive – it takes prices $\{p_i\}$ and wage w as given.

Step 1: Final Good Firm FOC (Inverse Demand)

The final good firm solves:

$$\max_{\{x_i\}, L_Y} \frac{L_Y^{1-\alpha}}{\alpha} \int_0^{M_t} x_i^\alpha \, di - \int_0^{M_t} p_i x_i \, di - w L_Y$$

FOC with respect to x_i:

$$L_Y^{1-\alpha} x_i^{\alpha-1} = p_i$$

This is the **inverse demand** facing intermediate producer i.

$$\boxed{p_i = L_Y^{1-\alpha} x_i^{\alpha-1}} \qquad \text{(Inverse Demand)}$$

Step 2: Intermediate Monopolist's Problem

Each intermediate firm produces x_i at marginal cost $c = \alpha$ (one unit of final good produces one unit of intermediate; the normalization sets $c = \alpha$).

The monopolist maximizes profit:

$$\pi_i = p_i(x_i)x_i - \alpha x_i = \left(L_Y^{1-\alpha} x_i^{\alpha-1}\right)x_i - \alpha x_i$$

FOC with respect to x_i:

$$\alpha L_Y^{1-\alpha} x_i^{\alpha-1} = \alpha \implies L_Y^{1-\alpha} x_i^{\alpha-1} = 1$$

Solving: $x_i = L_Y$ for all i (symmetric equilibrium).
The **monopoly price** is $p_i = 1$ (MC is α, markup is $1/\alpha$):

$$p_i = L_Y^{1-\alpha}(L_Y)^{\alpha-1} = L_Y^{1-\alpha} L_Y^{\alpha-1} = 1$$

Per-period profit:

$$\pi_i = (1-\alpha)x_i = (1-\alpha)L_Y = (1-\alpha)L$$

(since in equilibrium $L_Y = L$ when we ignore R&D labor for now, or $L_Y = L - L_R$).

$$\boxed{x_i = L_Y, \quad p_i = 1, \quad \pi_i = (1-\alpha)L_Y} \qquad \text{(Monopoly Solution)}$$

Step 3: Firm Value (Perpetual Monopoly)

With perpetual patent and constant interest rate r on the BGP:

$$V = \int_t^\infty \pi\, e^{-r(\tau - t)} d\tau = \frac{\pi}{r}$$

Step 4: R&D Sector and Free Entry

New varieties are created by an R&D sector. Assume the research technology uses resources R_t:

$$\dot{M}_t = \lambda R_t$$

Research firms are competitive. A researcher spending \$1 gets λ new blueprints, each worth $V = \pi/r$. Free entry drives profits to zero:

$$\pi^R = \frac{\pi}{r} \cdot \lambda - 1 = 0 \implies \boxed{r = \lambda\pi} \qquad \text{(Free Entry / Interest Rate)}$$

Step 5: Euler Equation and BGP Growth Rate

Households supply savings to R&D. Under CRRA utility, the Euler equation is:

$$\frac{\dot{c}}{c} = \frac{r - \rho}{\sigma}$$

On the BGP, $\gamma_{GDP} = \gamma_c$, and $\gamma_{GDP} = \dot{M}/M$ (since GDP grows proportionally to M).
Substituting $r = \lambda\pi = \lambda(1-\alpha)L_Y$:

$$\boxed{\gamma = \frac{\lambda(1-\alpha)L - \rho}{\sigma}} \qquad \text{(Romer Market Growth)}$$

(Setting $L_Y \approx L$ in the simplified lab-equipment version)

Aggregation on the BGP

With symmetric equilibrium $x_i = x = L$ and M_t varieties:

$$\begin{aligned}
Y_t &= \frac{M_t L}{\alpha} \\
X_t &= \int_0^{M_t} x_i \, di = LM_t \quad \text{(total intermediate usage)} \\
GDP_t &= Y_t - X_t = M_t L\Big(\frac{1}{\alpha} - 1\Big) = \frac{(1-\alpha)}{\alpha} M_t L
\end{aligned}$$

So $\gamma_{GDP} = \gamma_M = \dot{M}/M$.

6 Romer: Social Planner vs. Market (Distortions and Subsidy)

Source of Inefficiency in the Market

The market equilibrium is **inefficient** because:

- Intermediate monopolists charge $p_i = 1 > \alpha = MC$. The markup $1/\alpha$ restricts intermediate good usage below the social optimum.
- The planner sets $p_i = MC = \alpha$ (social marginal cost), and produces more of each variety.

Planner's Static Problem

Given M_t varieties, the planner maximizes net output by choosing x_i:

$$\max_{\{x_i\}} \frac{L^{1-\alpha}}{\alpha} \int_0^{M_t} x_i^{\alpha}\, di - \alpha \int_0^{M_t} x_i\, di$$

FOC:

$$L^{1-\alpha} x_i^{\alpha-1} = \alpha \implies x_i^* = L\alpha^{\frac{1}{\alpha-1}} > L = x_i^{mkt}$$

since $\alpha^{1/(\alpha-1)} > 1$ (because $\alpha \in (0,1)$ and the exponent $1/(\alpha-1) < 0$, so $\alpha^{1/(\alpha-1)} = 1/\alpha^{1/(1-\alpha)} > 1$).

$$x_i^* = L\alpha^{\frac{1}{\alpha-1}} > L = x_i^{mkt} \qquad \text{(Planner Static)}$$

Planner's Dynamic Problem and Growth Rate

Step 1: Compute planner's net output (using x_i^*):

$$\tilde{Y}_t^* = \frac{M_t}{\alpha} L\alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)$$

Step 2: Planner's Hamiltonian (present-value form):

$$\mathcal{J} = \frac{c_t^{1-\sigma} - 1}{1-\sigma} e^{-\rho t} + \nu_t \Big[\lambda \tilde{Y}_t^* - \lambda c_t\Big]$$

where the state variable is M_t and the law of motion is $\dot{M}_t = \lambda(\tilde{Y}_t^* - c_t)$.
Step 3: FOC with respect to c:

$$c_t^{-\sigma} e^{-\rho t} = \lambda \nu_t$$

Step 4: Costate equation (FOC w.r.t. M):

$$-\dot{\nu}_t = \nu_t \frac{\lambda L\alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} \implies \frac{\dot{\nu}}{\nu} = -\frac{\lambda L\alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha}$$

Step 5: Derive the planner Euler equation:

$$\frac{\dot{c}}{c} = -\frac{1}{\sigma}\Big(\frac{\dot{\nu}}{\nu} + \rho\Big) = \frac{1}{\sigma}\Big(\frac{\lambda L\alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} - \rho\Big)$$

$$\boxed{\gamma^* = \frac{1}{\sigma}\left(\frac{\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} - \rho\right)} \qquad \text{(Planner Growth)}$$

Market vs. Planner Comparison

Since $\alpha^{1/(\alpha-1)} > 1$, it follows that $\alpha^{\alpha/(\alpha-1)} = (\alpha^{1/(\alpha-1)})^\alpha > 1$, so:

$$\frac{\lambda L \alpha^{\frac{\alpha}{\alpha-1}}(1-\alpha)}{\alpha} > \lambda(1-\alpha)L$$

Therefore:

$$\gamma^* > \gamma$$

The planner's economy grows faster than the market because:

- The planner sets $p = MC = \alpha$ instead of $p = 1$ (monopoly price).
- More intensive use of intermediates raises output and R&D investment.
- The effective return to innovation is higher for the planner.

Optimal Policy: Subsidy on Intermediate Goods

Subsidize the use of intermediate goods so that intermediate firms face marginal cost α^2 and set price $p = \alpha^2/\alpha = \alpha =$ social marginal cost.
This requires a **subsidy per unit of** x_i equal to:

$$\boxed{\text{subsidy rate} = 1 - \alpha}$$

(So firms receive $1 - \alpha$ per unit sold, lowering the effective price from 1 to α.)

7 Scale Effects and Jones (1995) Fix

Scale Effects in the Romer Model

From the market growth rate: $\gamma = [\lambda(1-\alpha)L - \rho]/\sigma$.

The growth rate increases in L (total labor/population): **scale effects**.

Why? Research returns grow with market size: more workers $\Rightarrow$ larger market for innovations $\Rightarrow$ higher profits $\Rightarrow$ more R&D $\Rightarrow$ faster growth.

Empirical problem (Jones 1995): The number of US researchers increased more than 10× since the 1950s, but the growth rate of GDP did not accelerate.

Labor-Based Romer Model (First Attempt)

Replace lab equipment (resources) with labor as the research input:

$$\dot{M}_t = \lambda M_t L_t^R$$

Labor market clearing: $L = L_Y + L_R$.

Free entry into R&D: $\lambda M_t V = w_t$ (value of creating one new variety equals wage).

Derived growth rate:

$$\gamma = \frac{\alpha\lambda L - \rho}{\alpha + \sigma}$$

Result: Still has scale effects (γ increases in L).

Jones (1995): Semi-Endogenous Growth

Jones' fix: introduce **decreasing returns in blueprint production** (ideas get harder to find):

$$\dot{M}_t = \lambda M_t^{\phi} L_t^R, \qquad \phi < 1$$

With population growing at rate n, on the BGP:

Step 1: Researcher FOC (free entry):

$$L_E \lambda \alpha M^{\phi-1}/r = 1$$

Step 2: Take logs and differentiate with respect to time:

$$\dot{L}_E/L_E + (\phi - 1)\dot{M}/M = 0$$

Step 3: On the BGP, $L_E/L = \text{const}$, so $\dot{L}_E/L_E = n$.

Step 4: Substitute:

$$n + (\phi - 1)\frac{\dot{M}}{M} = 0 \implies \frac{\dot{M}}{M} = \frac{n}{1-\phi}$$

Since $\gamma_{GDP} = \gamma_M$:

$$\boxed{\gamma = \frac{n}{1-\phi}} \qquad \text{(Jones Semi-Endogenous)}$$

Key Conclusions from Jones (1995)

- Growth rate does **not** depend on L – no scale effects.
- Growth is driven entirely by population growth n.
- If $n = 0$: no sustained growth. The economy grows only in *levels* from knowledge spillovers, not in rates.
- Growth is **semi-endogenous**: endogenous in that it depends on ϕ and n, but the steady-state rate is pinned down by exogenous n.
- $\phi < 1$ means that doubling current research M makes each unit of research input less productive – decreasing returns to the stock of ideas.
- This is sometimes called the **fishing-out** effect: as more ideas are discovered, the remaining ideas are harder to find.

8 Comparative Analysis: Solow vs. AK vs. Romer

Feature	Solow	AK Model	Romer (Expanding Varieties)
Source of growth	Exogenous TFP (g)	Capital accumulation (no dim. returns)	Endogenous variety creation ($\dot{M}$)
Long-run growth driver	g (exogenous)	$sA - \delta$ (endogenous)	$\lambda(1-\alpha)L/\sigma$ (endogenous)
Convergence	Conditional convergence	No convergence	No convergence (BGP from start)
Transitional dynamics	Yes (to SS)	No	No
Effect of $\uparrow s$	Level effect only	Permanent growth effect	Growth effect (more R&D savings)
Scale effects	No	No	Yes ($\gamma \propto L$)
Role of savings	Determines k^* (level)	Determines growth rate	Determines R&D investment
Market structure	Perfect competition	Perfect competition	Monopolistic competition
Efficiency	Efficient (no externalities)	Inefficient (positive externality)	Inefficient (monopoly markup)
Returns to capital	Diminishing ($\alpha < 1$)	Constant ($\alpha = 1$)	N/A (no capital)
Key parameter	α (capital share)	A (productivity)	λ (research productivity), L

Summary of Growth Rates

Model	Per-capita growth rate	Depends on
Solow (SS)	g	Exogenous
AK (exogenous s)	$sA - \delta - n$	s, A, δ, n
AK-Ramsey	$(A - \delta - \rho)/\sigma$	A, δ, ρ, σ
Arrow-Frankel market	$(\alpha A - \delta - \rho)/\sigma$	$\alpha, A, \delta, \rho, \sigma$
Arrow-Frankel planner	$(A - \delta - \rho)/\sigma$	A, δ, ρ, σ
Romer market	$[\lambda(1-\alpha)L - \rho]/\sigma$	$\lambda, L, \alpha, \rho, \sigma$
Jones semi-endogenous	$n/(1-\phi)$	n, ϕ only

9 Problem-Solving Templates

(a) Derive the Growth Rate in the AK Model

1. Write the AK production function: $y = Ak$.
2. Write capital accumulation: $\dot{K} = sAK - \delta K$.
3. Divide by K: $\gamma_K = sA - \delta$.
4. Subtract population growth: $\gamma_k = sA - \delta - n$.
5. State the condition for positive per-capita growth: $sA > \delta + n$.
6. Note: this is immediate and constant – no transitional dynamics.

(b) Full Romer Equilibrium Sequence

1. **Final good FOC:** Differentiate Y w.r.t. x_i to get $p_i = L_Y^{1-\alpha} x_i^{\alpha-1}$.
2. **Intermediate firm:** Monopolist sets $\alpha L_Y^{1-\alpha} x_i^{\alpha-1} = c_i = \alpha$, giving $x_i = L_Y$.
3. **Monopoly price:** Substitute $x_i = L_Y$: $p_i = 1$ (or $p = 1/\alpha$ before normalization).
4. **Monopoly profit:** $\pi = (p - c)x = (1 - \alpha)L_Y$.
5. **Firm value:** $V = \pi/r$ (on BGP with constant r).
6. **Free entry:** Cost of creating 1 variety equals value: $1 = \lambda V \Rightarrow r = \lambda\pi$.
7. **Euler equation:** $\gamma_c = (r - \rho)/\sigma$.
8. **BGP:** $\gamma_c = \gamma_{GDP} = \gamma_M \Rightarrow$ combine to get $\gamma = [\lambda(1 - \alpha)L - \rho]/\sigma$.

(c) Market vs. Planner in Arrow-Frankel

1. **Market:** Firm j takes $\bar{A}$ as given. Private MPK $= \alpha\bar{A}K^{\alpha-1}$. Under AK case: $= \alpha A$. Market Euler: $\gamma_c = (\alpha A - \delta - \rho)/\sigma$.
2. **Planner:** Internalize $\bar{A} = A_0 K^{\eta}$. Social MPK $= (\alpha + \eta)\bar{A}K^{\alpha-1}$. Under AK case ($\eta = 1 - \alpha$): $= A$. Planner Euler: $\gamma_c^{SP} = (A - \delta - \rho)/\sigma$.
3. **Compare:** $\gamma_c^{SP} > \gamma_c^{mkt}$ since $A > \alpha A$.
4. **Subsidy:** Correct market to planner solution with subsidy $s^* = (1 - \alpha)/\alpha$ on capital.

(d) Jones (1995) BGP Derivation

1. Write research technology: $\dot{M} = \lambda M^{\phi} L_R$.
2. Write free entry (researcher FOC): $\Pi\lambda M^{\phi}/(rM) = w/L_E \Rightarrow$ after simplification: $L_E\lambda\alpha M^{\phi-1}/r = 1$.
3. Take logs and differentiate: $n + (\phi - 1)\gamma_M = 0$.
4. Solve: $\gamma_M = n/(1 - \phi) = \gamma_{GDP}$.
5. Conclude: no scale effects, growth proportional to n.

(e) Romer Market vs. Planner – Quick Comparison

1. Market: $x_i = L$, $p_i = 1$, $\pi = (1-\alpha)L$, $r = \lambda(1-\alpha)L$, $\gamma = [\lambda(1-\alpha)L - \rho]/\sigma$.
2. Planner: $x_i^* = L\alpha^{1/(\alpha-1)} > L$, higher output, $\gamma^* > \gamma$.
3. Correction: subsidy $(1-\alpha)$ per unit of x_i to make effective price equal to social MC.

10 Common Pitfalls

1. **Confusing AK with Solow transitional dynamics.**
   - In Solow, the economy has diminishing returns and transitions to a unique steady state.
   - In the AK model, there is *no* steady state in the usual sense – the economy always grows at rate $sA - \delta$. Do not draw a phase diagram with a crossing point for AK.
2. **Confusing private and social returns in Arrow-Frankel.**
   - Private MPK $= \alpha A$ (firm takes $\bar{A}$ as given).
   - Social MPK $= A$ (planner internalizes $\bar{A} = A_0 K^\eta$ with $\eta = 1 - \alpha$).
   - This is the key wedge. Stating that both equal A is a common error.
3. **Scale effects in Romer.**
   - The market growth rate $\gamma = [\lambda(1-\alpha)L - \rho]/\sigma$ increases in L.
   - Jones (1995) documents that this contradicts the data (R&D workers grew, growth did not).
   - The fix is $\phi < 1$ in $\dot{M} = \lambda M^\phi L_R$, which eliminates scale effects.
4. **Forgetting the monopoly markup in Romer.**
   - Intermediate goods are priced at $p = 1$, not at MC $= \alpha$.
   - The markup is $1/\alpha$. This is the source of inefficiency in the market allocation.
   - The firm's optimal quantity is $x_i = L_Y$ (not smaller), but the *price* is too high.
5. **Wrong expression for V outside the BGP.**
   - $V = \pi/r$ holds only on the BGP where r is constant.
   - Off-BGP, $V = \int_t^\infty \pi e^{-\int_t^\tau r_\omega d\omega} d\tau$.
6. **Confusing Y with GDP in the Romer model.**
   - Y_t is gross output of the final good. $GDP_t = Y_t - X_t$ subtracts intermediate goods cost.
   - Growth of GDP equals growth of M (on the symmetric BGP).
7. **Misapplying the AK-Ramsey Euler equation.**
   - The Euler equation is $\gamma_c = (A - \delta - \rho)/\sigma$.
   - This requires $A > \delta + \rho$; otherwise consumption declines ($\gamma_c < 0$), which violates the TVC.
   - The growth rate is *constant*, not time-varying, because $r = A - \delta$ is constant.
8. **Jones semi-endogenous and the role of n.**
   - If population is constant ($n = 0$), there is *no growth* in the Jones model.
   - Growth requires $n > 0$; the growth rate is $n/(1-\phi)$.
   - Do not confuse the level of L (which does not affect growth) with its *growth rate* n (which does).

11 Quick Reference Formulas

AK Model (Exogenous Savings)

$$y = Ak \quad \text{(production)}$$
$$\gamma_k = \gamma_y = sA - \delta - n \quad \text{(growth rate per capita)}$$
$$\text{Positive growth iff } sA > \delta + n$$
$$\text{No transitional dynamics; no convergence}$$

AK-Ramsey (Endogenous Savings)

$$\dot{k} = (A - \delta - n)k - c \quad \text{(resource constraint)}$$
$$H^{CV} = \frac{c^{1-\sigma}}{1-\sigma} + \lambda\big[(A - \delta - n)k - c\big] \quad \text{(Hamiltonian)}$$
$$\gamma_c = \frac{A - \delta - \rho}{\sigma} \quad \text{(Euler equation)}$$
$$\text{TVC:} \quad \rho > (1-\sigma)\gamma_c$$

Arrow-Frankel Learning-by-Doing

$$Y_j = A_0 K^{\eta} K_j^{\alpha} L_j^{1-\alpha} \quad \text{(firm, AK case: } \eta = 1 - \alpha\text{)}$$
$$Y = A_0 K^{\eta+\alpha} L^{1-\alpha} = AK \quad \text{(aggregate, AK case with } L = 1\text{)}$$
$$\gamma_c^{mkt} = \frac{\alpha A - \delta - \rho}{\sigma} \quad \text{(market, uses private MPK)}$$
$$\gamma_c^{SP} = \frac{A - \delta - \rho}{\sigma} \quad \text{(planner, uses social MPK)}$$
$$s^* = \frac{1-\alpha}{\alpha} \quad \text{(optimal capital subsidy)}$$

Romer Expanding Varieties

$$Y_t = \frac{L_Y^{1-\alpha}}{\alpha} \int_0^{M_t} x_i^{\alpha}\, di \quad \text{(final good)}$$
$$p_i = L_Y^{1-\alpha} x_i^{\alpha-1} \quad \text{(inverse demand)}$$
$$x_i = L_Y, \quad p_i = 1, \quad \pi = (1-\alpha)L_Y \quad \text{(monopoly solution)}$$
$$V = \frac{\pi}{r} \quad \text{(firm value on BGP)}$$
$$r = \lambda\pi \quad \text{(free entry)}$$
$$\gamma = \frac{\lambda(1-\alpha)L - \rho}{\sigma} \quad \text{(BGP growth rate, market)}$$
$$\gamma^* > \gamma \quad \text{(planner grows faster)}$$

Jones (1995) Semi-Endogenous

$$\dot{M}_t = \lambda M_t^{\phi} L_t^{R}, \quad \phi < 1$$
$$\gamma = \frac{n}{1-\phi} \quad \text{(BGP growth rate)}$$
$$\text{No scale effects; } n = 0 \Rightarrow \text{no growth}$$

GDP in Romer (Symmetric BGP)

$$Y_t = \frac{M_t L}{\alpha}, \quad GDP_t = \frac{(1-\alpha)}{\alpha} M_t L, \quad \gamma_{GDP} = \gamma_M$$